

北京邮电大学

《网络安全》课程实验报告

实验名称： 实验 3：网络扫描实验

学生姓名：

学 号：

班 级：

实验 3：网络扫描课堂实验

一、实验目的

用 Nmap 对主机进行主机扫描、ping 扫描、端口扫描、TCP connect 扫描、操作系统识别、漏洞扫描等，获取尽可能全面的主机信息，抓包分析扫描原理；

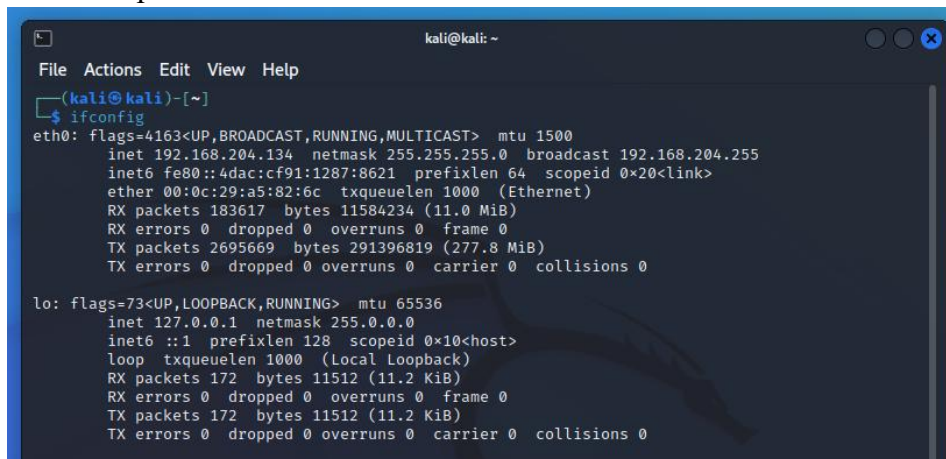
二、实验环境

- (1) 扫描机：Kali，配置 nmap 工具；
- (2) 靶机：Windows XP；
- (3) 软件：Nmap、wireshark

三、实验过程与结果

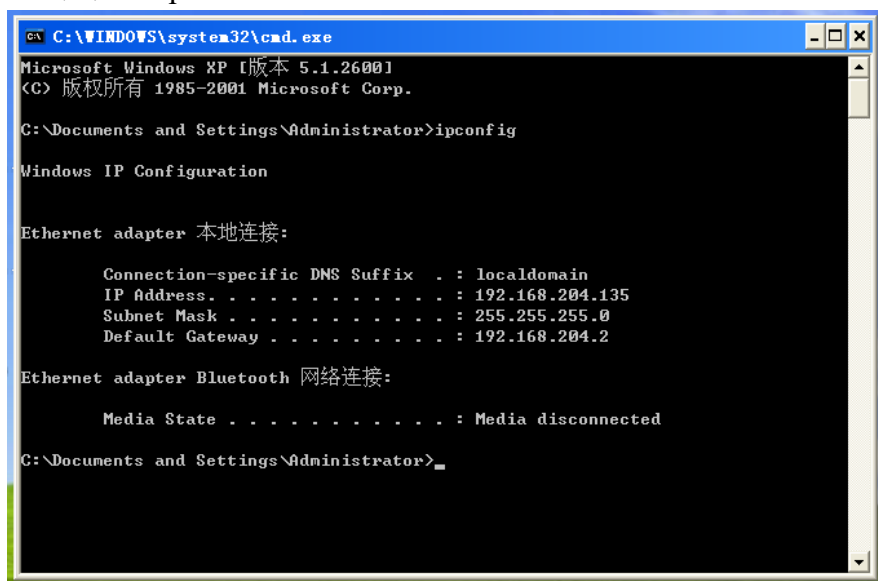
3.1 步骤一：查看主机信息

1.在扫描机 Kali 中打开终端，输入“ifconfig”命令查看扫描机信息，如下图所示可知扫描机的 ip 地址为 192.168.204.134。



```
kali@kali: ~  
File Actions Edit View Help  
$ ifconfig  
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet 192.168.204.134 netmask 255.255.255.0 broadcast 192.168.204.255  
    inet6 fe80::4dac:cf91:1287:8621 prefixlen 64 scopeid 0x20<link>  
    ether 00:0c:29:a5:82:6c txqueuelen 1000 (Ethernet)  
    RX packets 183617 bytes 11584234 (11.0 MiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 2695669 bytes 291396819 (277.8 MiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536  
    inet 127.0.0.1 netmask 255.0.0.0  
    inet6 ::1 prefixlen 128 scopeid 0x10<host>  
    loop txqueuelen 1000 (Local Loopback)  
    RX packets 172 bytes 11512 (11.2 KiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 172 bytes 11512 (11.2 KiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

2.在靶机 windows XP 中打开命令行，使用“ipconfig”命令查看靶机信息，如下图所示可知靶机的 ip 地址为 192.168.204.135。



```
C:\WINDOWS\system32\cmd.exe  
Microsoft Windows XP [版本 5.1.2600]  
<C> 版权所有 1985-2001 Microsoft Corp.  
  
C:\Documents and Settings\Administrator>ipconfig  
  
Windows IP Configuration  
  
Ethernet adapter 本地连接:  
  
    Connection-specific DNS Suffix  . : localdomain  
    IP Address. . . . . : 192.168.204.135  
    Subnet Mask . . . . . : 255.255.255.0  
    Default Gateway . . . . . : 192.168.204.2  
  
Ethernet adapter Bluetooth 网络连接:  
  
    Media State . . . . . : Media disconnected  
  
C:\Documents and Settings\Administrator>
```

3.在扫描机 kali 终端中使用 `ping 192.168.204.135` 命令 ping 靶机，按“ctrl+c”停止，如下图所示，收到来自靶机的答复，说明扫描机 ping 通靶机。

```
(kali㉿kali)-[~]
$ ping 192.168.204.135
PING 192.168.204.135 (192.168.204.135) 56(84) bytes of data:
 64 bytes from 192.168.204.135: icmp_seq=1 ttl=128 time=3.50 ms
 64 bytes from 192.168.204.135: icmp_seq=2 ttl=128 time=1.65 ms
 64 bytes from 192.168.204.135: icmp_seq=3 ttl=128 time=1.39 ms
 64 bytes from 192.168.204.135: icmp_seq=4 ttl=128 time=1.11 ms
 64 bytes from 192.168.204.135: icmp_seq=5 ttl=128 time=2.11 ms
 64 bytes from 192.168.204.135: icmp_seq=6 ttl=128 time=3.46 ms
 64 bytes from 192.168.204.135: icmp_seq=7 ttl=128 time=1.69 ms
 64 bytes from 192.168.204.135: icmp_seq=8 ttl=128 time=1.24 ms
 64 bytes from 192.168.204.135: icmp_seq=9 ttl=128 time=1.71 ms
 64 bytes from 192.168.204.135: icmp_seq=10 ttl=128 time=1.50 ms
 64 bytes from 192.168.204.135: icmp_seq=11 ttl=128 time=2.14 ms
 64 bytes from 192.168.204.135: icmp_seq=12 ttl=128 time=2.34 ms
 64 bytes from 192.168.204.135: icmp_seq=13 ttl=128 time=1.91 ms
 64 bytes from 192.168.204.135: icmp_seq=14 ttl=128 time=1.25 ms
 64 bytes from 192.168.204.135: icmp_seq=15 ttl=128 time=1.70 ms
 64 bytes from 192.168.204.135: icmp_seq=16 ttl=128 time=1.35 ms
 64 bytes from 192.168.204.135: icmp_seq=17 ttl=128 time=1.47 ms
 64 bytes from 192.168.204.135: icmp_seq=18 ttl=128 time=1.27 ms
 64 bytes from 192.168.204.135: icmp_seq=19 ttl=128 time=1.52 ms
 64 bytes from 192.168.204.135: icmp_seq=20 ttl=128 time=1.53 ms
 64 bytes from 192.168.204.135: icmp_seq=21 ttl=128 time=1.56 ms
 64 bytes from 192.168.204.135: icmp_seq=22 ttl=128 time=1.47 ms
 64 bytes from 192.168.204.135: icmp_seq=23 ttl=128 time=1.37 ms
 64 bytes from 192.168.204.135: icmp_seq=24 ttl=128 time=1.50 ms
 64 bytes from 192.168.204.135: icmp_seq=25 ttl=128 time=2.34 ms
 64 bytes from 192.168.204.135: icmp_seq=26 ttl=128 time=1.34 ms
 64 bytes from 192.168.204.135: icmp_seq=27 ttl=128 time=2.98 ms
^C
— 192.168.204.135 ping statistics —
27 packets transmitted, 27 received, 0% packet loss, time 2605ms
rtt min/avg/max/mdev = 1.110/1.793/3.502/0.627 ms
```

4.在靶机 windows XP 命令行中使用 `ping 192.168.204.134` 命令 ping 扫描机，如下图所示，收到来自扫描机的答复，说明靶机 ping 通扫描机。至此，两主机实现 ping 通。

```
C:\Documents and Settings\Administrator>ping 192.168.204.134

Pinging 192.168.204.134 with 32 bytes of data:

Reply from 192.168.204.134: bytes=32 time=1ms TTL=64
Reply from 192.168.204.134: bytes=32 time=1ms TTL=64
Reply from 192.168.204.134: bytes=32 time=1ms TTL=64
Reply from 192.168.204.134: bytes=32 time=2ms TTL=64

Ping statistics for 192.168.204.134:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

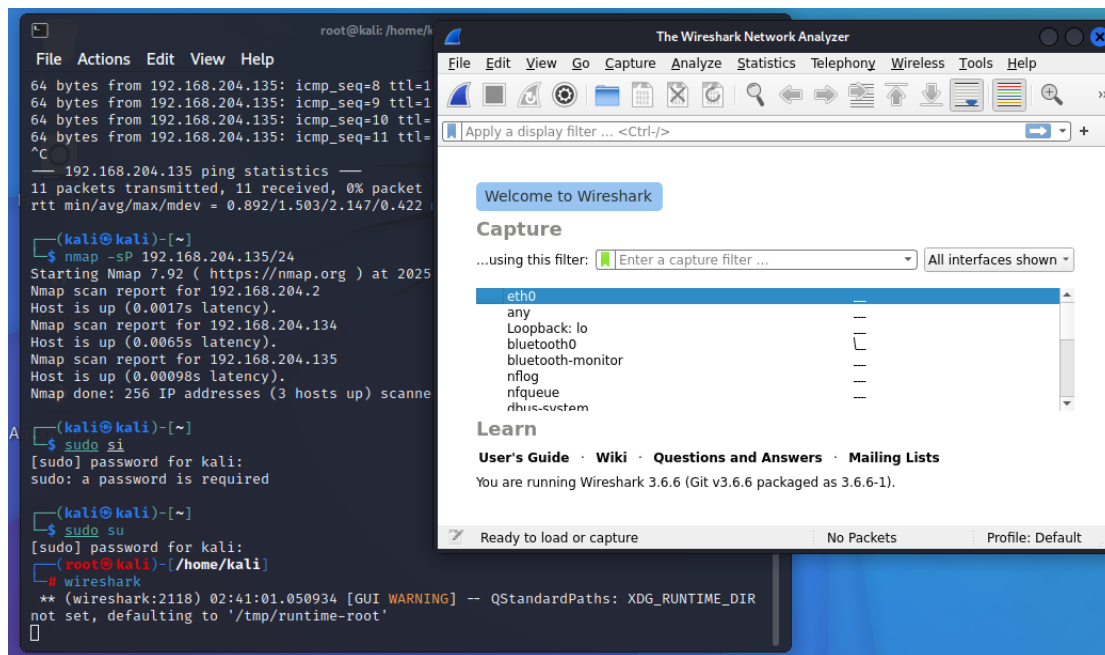
C:\Documents and Settings\Administrator>
```

3.2 步骤二：Nmap Ping 扫描

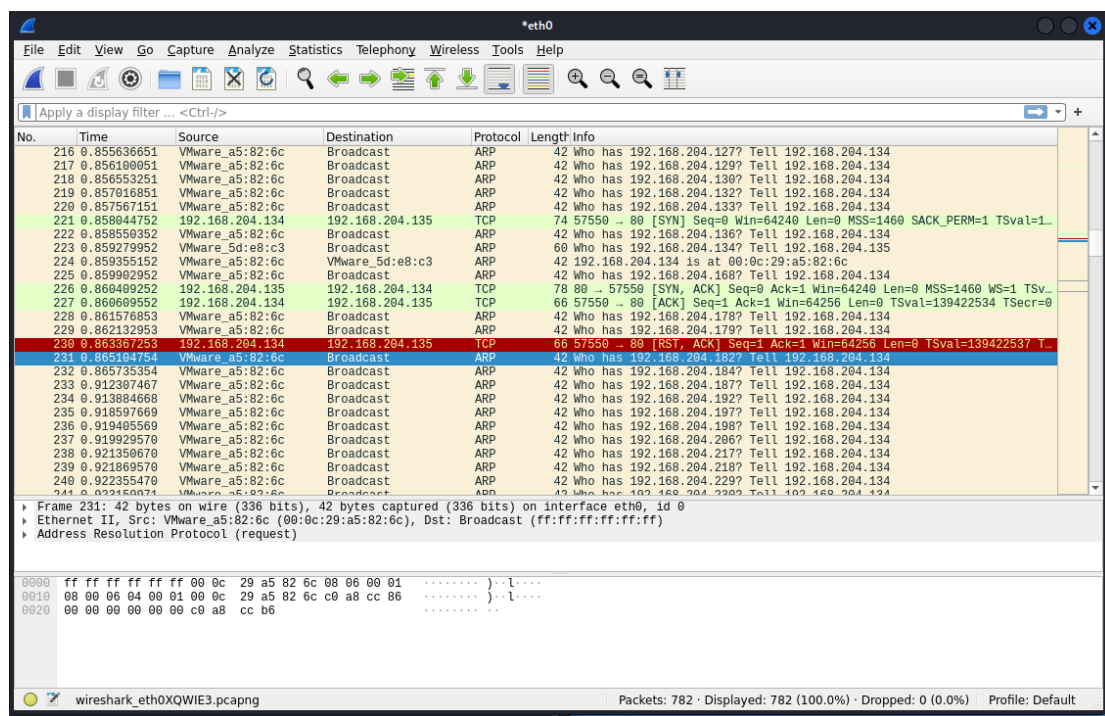
1.在扫描机终端中使用 `nmap -sP 192.168.204.135/24` 命令，对靶机进行 Ping 扫描，收集 C 段内部属于目标的 IP 地址。如下图终端输出的扫描结果可知，成功扫描到网关 192.168.204.2、靶机 192.168.204.135、扫描机 192.168.204.134。

```
(kali㉿kali)-[~]
$ nmap -sP 192.168.204.135/24
Starting Nmap 7.92 ( https://nmap.org ) at 2025-10-30 02:39 EDT
Nmap scan report for 192.168.204.2
Host is up (0.0017s latency).
Nmap scan report for 192.168.204.134
Host is up (0.0065s latency).
Nmap scan report for 192.168.204.135
Host is up (0.00098s latency).
Nmap done: 256 IP addresses (3 hosts up) scanned in 2.60 seconds
```

2.在扫描机 kali 终端输入 `sudo su` 获取超级用户权限，接着输入 `wireshark`，在超级用户权限下打开 wireshark。



3.进行扫描的同时打开 wireshark 捕获报文，捕获结果如下图所示。



4.在上图中可以看到，wireshark 在此次主机扫描过程中捕获到若干 TCP 包、DNS 包和大量由物理地址 00:0c:29:a5:82:6c 广播的 ARP 包。

(1) 前三个 TCP 包：由扫描机 kali 向 IP 地址 192.168.204.1 和 198.168.204.2 的 80 端口均发出 TCP 请求 SYN，稍后由 192.168.204.2 回复给扫描机[RST,ACK]。说明此 IP 地址对应网关，且网关的 80 端口未打开。

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	192.168.204.134	192.168.204.1	TCP	74	54750 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=2...
2	0.000474400	192.168.204.134	192.168.204.2	TCP	74	51632 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=1...
3	0.000790300	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.3? Tell 192.168.204.134
4	0.001167900	192.168.204.2	192.168.204.134	TCP	60	80 → 51632 [RST, ACK] Seq=1 Ack=1 Win=32767 Len=0
5	0.001200400	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.4? Tell 192.168.204.134
6	0.001634100	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.5? Tell 192.168.204.134
7	0.002091201	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.6? Tell 192.168.204.134

(2) 由扫描机 kali 广播发送若干 ARP 包：扫描机依次对 IP 地址 192.168.204.3-253 进行查询，要求查询结果发送回扫描机 192.168.204.134。

*eth0						
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help						
Apply a display filter ... <Ctrl-/>						
No.	Time	Source	Destination	Protocol	Length	Info
128	0.726262513	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.212? Tell 192.168.204.134
129	0.727036113	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.213? Tell 192.168.204.134
130	0.727597913	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.214? Tell 192.168.204.134
131	0.728288014	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.215? Tell 192.168.204.134
132	0.728595114	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.216? Tell 192.168.204.134
133	0.729321114	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.219? Tell 192.168.204.134
134	0.730438814	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.220? Tell 192.168.204.134
135	0.731093415	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.221? Tell 192.168.204.134
136	0.731523015	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.222? Tell 192.168.204.134
137	0.731949115	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.223? Tell 192.168.204.134
138	0.732581015	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.224? Tell 192.168.204.134
139	0.733082215	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.225? Tell 192.168.204.134
140	0.733623615	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.226? Tell 192.168.204.134
141	0.734190515	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.227? Tell 192.168.204.134
142	0.734689415	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.228? Tell 192.168.204.134
143	0.736211616	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.231? Tell 192.168.204.134
144	0.736799816	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.232? Tell 192.168.204.134
145	0.737302316	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.233? Tell 192.168.204.134
146	0.737811316	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.234? Tell 192.168.204.134
147	0.738385516	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.235? Tell 192.168.204.134
148	0.739193817	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.236? Tell 192.168.204.134
149	0.739654617	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.237? Tell 192.168.204.134
150	0.740196617	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.238? Tell 192.168.204.134
151	0.811270938	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.241? Tell 192.168.204.134
152	0.812324638	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.244? Tell 192.168.204.134
153	0.915756139	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.247? Tell 192.168.204.134
Frame 132: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface eth0, id 0						
Ethernet II, Src: VMware_a5:82:6c (00:0c:29:a5:82:6c), Dst: Broadcast (ff:ff:ff:ff:ff:ff)						
Address Resolution Protocol (request)						

(3) 靶机发送 ARP 包：当扫描机发出查询 IP 地址 192.168.117.133 的 ARP 包后，靶机向扫描机发送 ARP 包，告知扫描机其 IP 地址所对应的 Mac 地址为 00:0c:29:a5:82:6c。

(4) 扫描机和靶机间交换 TCP 包：扫描机向靶机 80 端口发送 TCP 建立连接请求，稍后由靶机 80 端口返回[RST,ACK]，说明此主机是开启的。

152	0.812324638	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.244? Tell 192.168.204.134
153	0.815766139	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.247? Tell 192.168.204.134
154	0.817017949	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.250? Tell 192.168.204.134
155	0.817501349	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.251? Tell 192.168.204.134
156	0.818697549	192.168.204.134	192.168.204.254	TCP	74	45450 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=3...
157	0.819578949	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.0? Tell 192.168.204.134
158	0.820202941	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.11? Tell 192.168.204.134
159	0.820669249	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.12? Tell 192.168.204.134
218	0.856553251	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.130? Tell 192.168.204.134
219	0.857016851	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.132? Tell 192.168.204.134
220	0.857567151	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.133? Tell 192.168.204.134
221	0.858094732	192.168.204.134	192.168.204.135	TCP	74	57550 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=1...
222	0.858550352	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.136? Tell 192.168.204.134
223	0.859279952	VMware_5d:e8:c3	Broadcast	ARP	60	Who has 192.168.204.134? Tell 192.168.204.135
224	0.859355152	VMware_a5:82:6c	VMware_5d:e8:c3	ARP	42	192.168.204.134 is at 00:0c:29:a5:82:6c
225	0.859902952	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.168? Tell 192.168.204.134
226	0.860409252	192.168.204.135	192.168.204.134	TCP	78	80 → 57550 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 WS=1 TSv...
227	0.860609552	192.168.204.134	192.168.204.135	TCP	66	57550 → 80 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=139422534 TSecr=0
228	0.861576853	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.178? Tell 192.168.204.134
229	0.862132953	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.179? Tell 192.168.204.134
230	0.863367253	192.168.204.134	192.168.204.135	TCP	66	57550 → 80 [RST, ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=139422537 T...
231	0.865104754	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.182? Tell 192.168.204.134
232	0.865735354	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.184? Tell 192.168.204.134
233	0.912307467	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.187? Tell 192.168.204.134
270	1.025247600	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.193? Tell 192.168.204.134
271	1.104447824	192.168.204.134	192.168.204.1	TCP	74	54762 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=2...
272	1.107144425	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.21? Tell 192.168.204.134

(5) 扫描机和网关间交换 DNS 包：在发送 ARP 包对 c 段各 IP 地址都进行了一遍查询之后，扫描机 192.168.204.134 和网关 192.168.204.2 之间交换 DNS 报文进行域名解析。

562	2.358115092	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.71? Tell 192.168.204.134
563	2.358115092	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.71? Tell 192.168.204.134
564	2.440694918	192.168.204.134	192.168.204.2	DNS	86	Standard query 0x2d30 PTR 2.204.168.192.in-addr.arpa
565	2.450588218	192.168.204.134	192.168.204.2	DNS	88	Standard query 0x2d31 PTR 134.204.168.192.in-addr.arpa
566	2.452182819	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.74? Tell 192.168.204.134
567	2.452211719	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.77? Tell 192.168.204.134
568	2.452236019	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.80? Tell 192.168.204.134
569	2.452259119	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.83? Tell 192.168.204.134
570	2.452283319	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.86? Tell 192.168.204.134
571	2.452306219	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.89? Tell 192.168.204.134
572	2.452329919	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.90? Tell 192.168.204.134
573	2.452369519	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.93? Tell 192.168.204.134
574	2.452413819	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.94? Tell 192.168.204.134
575	2.452481819	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.95? Tell 192.168.204.134
576	2.452526819	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.98? Tell 192.168.204.134
577	2.453080319	192.168.204.134	192.168.204.2	DNS	88	Standard query 0x2d32 PTR 135.204.168.192.in-addr.arpa
578	2.453090921	192.168.204.2	192.168.204.134	DNS	143	Standard query response 0x2d31 No such name PTR 134.204.168.192.in-...
579	2.453739523	192.168.204.2	192.168.204.134	DNS	141	Standard query response 0x2d30 No such name PTR 2.204.168.192.in-ad...
580	2.471074224	192.168.204.2	192.168.204.134	DNS	143	Standard query response 0x2d32 No such name PTR 135.204.168.192.in-...
581	2.547097177	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.101? Tell 192.168.204.134
582	2.547306447	VMware_a5:82:6c	Broadcast	ARP	42	Who has 192.168.204.103? Tell 192.168.204.134

5.由以上分析可知，Nmap Ping 扫描的原理是通过发送 ARP 包对 c 段所有 IP 进行查询，主机回复 ARP 包告知其 Mac 地址后，对查询到的主机 80 端口进行 TCP 连接请求，若收到回复则确认该主机开启。最终终端查询结果输出全部查询到的主机 IP 地址。

3.3 步骤三：Nmap 主机扫描

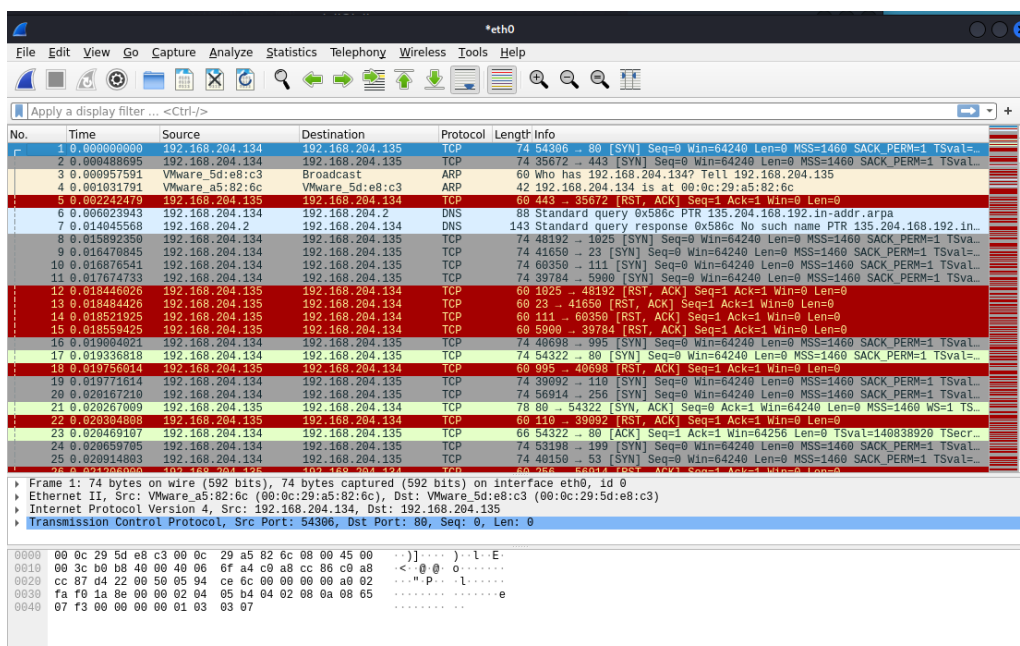
1.使用 `nmap -A 192.168.204.135` 命令对靶机 Windows XP 进行主机扫描。

```
(kali㉿kali)-[~]
$ nmap -A 192.168.204.135
Starting Nmap 7.92 ( https://nmap.org ) at 2025-10-30 02:42 EDT
Nmap scan report for 192.168.204.135
Host is up (0.0047s latency).
Not shown: 997 closed tcp ports (conn-refused)
PORT      STATE SERVICE        VERSION
135/tcp    open  msrpc          Microsoft Windows RPC
139/tcp    open  netbios-ssn    Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds   Windows XP microsoft-ds
Service Info: OSs: Windows, Windows XP; CPE: cpe:/o:microsoft:windows, cpe:/o:microsoft:windows_xp

Host script results:
|_clock-skew: mean: -4h00m00s, deviation: 5h39m24s, median: -8h00m00s
|_smb2-time: Protocol negotiation failed (SMB2)
|_nbstat: NetBIOS name: WINXP-3D8CC5A43, NetBIOS user: <unknown>, NetBIOS MAC: 00:0c:29:5d:e8:c3 (VMware)
|_smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_ message_signing: disabled (dangerous, but default)
|_smb-os-discovery:
|   OS: Windows XP (Windows 2000 LAN Manager)
|   OS CPE: cpe:/o:microsoft:windows_xp::-
|   Computer name: winxp-3d8cc5a43
|   NetBIOS computer name: WINXP-3D8CC5A43\x00
|   Workgroup: WORKGROUP\x00
|_ System time: 2025-10-30T14:42:10+08:00

Service detection performed. Please report any incorrect results at https://nmap.org/s
ubmit/ .
Nmap done: 1 IP address (1 host up) scanned in 18.99 seconds
```

2.对靶机主机进行扫描的同时打开 wireshark 捕获报文，捕获结果如下图所示。



3.由上图可知，捕获到两个 ARP 包、若干 TCP 包和若干 DNS 协议包。

(1) ARP 包：首先由 Mac 地址为 00:0c:29:a5:82:6c 的扫描机广播发出 ARP 包，查询扫描目标 IP 地址所对应的 Mac 地址，靶机收到后向扫描机回复 ARP 包说明其 Mac 地址为 00:0c:29:5d:e8:c3。

3	0.000957591	VMware_5d:e8:c3	Broadcast	ARP	60 Who has 192.168.204.134? Tell 192.168.204.135
4	0.001031791	VMware_a5:82:6c	VMware_5d:e8:c3	ARP	42 192.168.204.134 is at 00:0c:29:a5:82:6c
5	0.002242479	192.168.204.135	192.168.204.134	TCP	60 443 → 35672 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
6	0.006023943	192.168.204.134	192.168.204.2	DNS	88 Standard query 0x586c PTR 135.204.168.192.in-addr.arpa

(2) TCP 包：查看发送方地址 Source 和接收方地址 Destination 可知，这些 TCP 包在扫描机 kali 和靶机 windows XP 之间传送。

a) 首先由扫描机发出 TCP 请求包，分别向靶机 80 和 443 端口发送 SYN，这两个端口是常用的 HTTP 和 HTTPS 服务端口。

b) 靶机 80 和 443 端口均回复扫描机[RST,ACK]拒绝建立连接，说明主机在线。

4	0.001031791	VMware_a5:82:6c	VMware_5d:e8:c3	ARP	42 192.168.204.134 is at 00:0c:29:a5:82:6c
5	0.002242479	192.168.204.135	192.168.204.134	TCP	60 443 → 35672 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
6	0.006023943	192.168.204.134	192.168.204.2	DNS	88 Standard query 0x586c PTR 135.204.168.192.in-addr.arpa

3. DNS 包：同主机扫描中的分析，扫描机 192.168.204.134 和网关 192.168.204.2 交换 DNS 协议包进行域名解析。

6	0.006023943	192.168.204.134	192.168.204.2	DNS	88 Standard query 0x586c PTR 135.204.168.192.in-addr.arpa
7	0.014045568	192.168.204.2	192.168.204.134	DNS	143 Standard query response 0x586c No such name PTR 135.204.168.192.in-addr.arpa

4.由以上分析可知，Nmap 主机扫描的原理是发送 ARP 包确定扫描目的 IP 所对应的 Mac 地址后，向目标 IP 的 80 和 443 端口发送建立 TCP 连接请求，收到 TCP 应答报文则说明主机在线。

5.由 Ping 扫描和主机扫描分析的共同点可知，每次对靶机进行扫描前都需要扫描机发送 ARP 包查询靶机 Mac 地址，收到靶机 ARP 回复后开始扫描；而每次扫描也都需要扫描机与网关交换 DNS 报文进行域名解析。这些共同点在接下来的扫描原理分析中不再赘述。

3.4 步骤四：端口扫描

3.4.1 -sV：探测开放端口以确定服务/版本信息

1.在扫描机 kali 终端使用 **nmap -sV 192.168.204.135** 命令对靶机进行端口扫描，在终端输出的扫描结果如下图所示。

```
(kali㉿kali)-[~]
└─$ nmap -sV 192.168.204.135
Starting Nmap 7.92 ( https://nmap.org ) at 2025-10-30 02:43 EDT
Nmap scan report for 192.168.204.135
Host is up (0.0042s latency).
Not shown: 997 closed tcp ports (conn-refused)
PORT      STATE SERVICE        VERSION
135/tcp   open  msrpc          Microsoft Windows RPC
139/tcp   open  netbios-ssn    Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds   Microsoft Windows XP microsoft-ds
Service Info: OSs: Windows, Windows XP; CPE: cpe:/o:microsoft:windows, cpe:/o:microsoft:windows_xp

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.78 seconds
```

2.扫描结果显示，靶机所开启的端口有 135、139、445 端口，并列出了端口对应的服务和版本。

3.验证靶机以上端口打开：在靶机命令行中输入 **telnet 192.168.204.135+端口号** 检测对应端口是否打开，如下两图所示，端口号为 80 时输出“连接失败”说明靶机 80 端口未开放；而端口号为 135 时跳转到“Telnet 192.168.204.135”界面说明靶机 135 端口已开放，端口 139、445 同理。

```
(kali㉿kali)-[~]
└─$ telnet 192.168.204.135 80
Trying 192.168.204.135 ...
telnet: Unable to connect to remote host: Connection refused
```

```
(kali㉿kali)-[~]
└─$ telnet 192.168.204.135 135
Trying 192.168.204.135 ...
Connected to 192.168.204.135.
Escape character is '^['.
```

4.在进行上述端口扫描的同时使用 **wireshark** 捕获网络流量，如下图所示。

No.	Time	Source	Destination	Protocol	Length	Info
17	0.061391914	192.168.204.134	192.168.204.135	TCP	74	58146 → 3389 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=...
18	0.062405109	192.168.204.135	192.168.204.134	TCP	60	3389 → 58146 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
19	0.063667203	192.168.204.134	192.168.204.135	TCP	74	41980 → 1025 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=...
20	0.064481800	192.168.204.134	192.168.204.135	TCP	74	51586 → 111 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=...
21	0.065300596	192.168.204.135	192.168.204.134	TCP	60	1025 → 41980 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
22	0.065874593	192.168.204.135	192.168.204.134	TCP	60	111 → 51586 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
23	0.067292487	192.168.204.134	192.168.204.135	TCP	74	41232 → 3306 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=...
24	0.068171982	192.168.204.134	192.168.204.135	TCP	74	41032 → 587 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=...
25	0.068623280	192.168.204.135	192.168.204.134	TCP	60	3306 → 41232 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
26	0.068663180	192.168.204.135	192.168.204.134	TCP	60	587 → 41032 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
27	0.070399212	192.168.204.134	192.168.204.135	TCP	74	40994 → 5900 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=...
28	0.072924164	192.168.204.135	192.168.204.134	TCP	60	5900 → 40984 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
29	0.072331463	192.168.204.134	192.168.204.135	TCP	74	45032 → 8080 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=...
30	0.073563057	192.168.204.135	192.168.204.134	TCP	60	8080 → 45032 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
31	0.074267354	192.168.204.134	192.168.204.135	TCP	66	33136 → 135 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0 TSval=141207682...
32	0.075664347	192.168.204.134	192.168.204.135	TCP	74	50360 → 23 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=...
33	0.076564743	192.168.204.134	192.168.204.135	TCP	74	36880 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=...
34	0.077268240	192.168.204.135	192.168.204.134	TCP	60	23 → 50360 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
35	0.078043236	192.168.204.135	192.168.204.134	TCP	60	443 → 36880 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
36	0.078612634	192.168.204.134	192.168.204.135	TCP	74	41114 → 256 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=...
37	0.079314230	192.168.204.134	192.168.204.135	TCP	74	50166 → 110 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=...
38	0.079750329	192.168.204.135	192.168.204.134	TCP	60	256 → 41114 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
39	0.079750928	192.168.204.135	192.168.204.134	TCP	60	110 → 50166 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
40	0.083323512	151.101.109.91	192.168.204.134	TLSv1.2	173	Application Data
41	0.083455511	192.168.204.134	151.101.109.91	TCP	54	49544 → 443 [ACK] Seq=147 Ack=120 Win=63784 Len=0
42	0.084604166	192.168.204.134	192.168.204.135	TCP	74	45528 → 1792 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=...
▶ Frame 1: 200 bytes on wire (1600 bits), 200 bytes captured (1600 bits) on interface eth0, id 0 ▶ Ethernet II, Src: VMware_a5:82:6c (00:0c:29:a5:82:6c), Dst: VMware_e8:44:a5 (00:50:56:e8:44:a5) ▶ Internet Protocol Version 4, Src: 192.168.204.134, Dst: 151.101.109.91 ▶ Transmission Control Protocol, Src Port: 49544, Dst Port: 443, Seq: 1, Ack: 1, Len: 146 ▶ Transport Layer Security						
0000	00 50 56 e8 44 a5 00 0c	29 a5 82 6c 08 00 45 00	PV D...	...	l..E..	
0010	00 ba db 49 40 00 40 06	cd 04 c0 a8 cc 86 97 65	...I@...	...	e	
0020	6d 5b 01 88 01 bb 3c 3e	69 c6 5d b6 73 b1 50 18	m[...<	...	1.]s.P	
0030	f9 28 92 9c 00 00 17 03	03 00 0d 55 e0 3a 1e 31	{.....	...	U...:1	
0040	a0 50 d1 4d 53 cc a4 af	f0 2d c0 5d 30 4d 96 23	XMS...	...	..]0M...	
0050	29 b5 d4 dd c7 62 77 b7	7d e3 f7 cc 09 20 9f 8f	bw...	...		
0060	d5 34 ef 78 83 8d 45 3b	e4 1c 23 6a ac f7 1b d4	.4x..E;	...	..#j....	
0070	84 b9 18 cc e2 65 d3 ae	2f 71 9d ac 6d 5c d4 4f	e...	...	/q..m\0	

5.由上图可知，除了用于查询靶机和网关 Mac 地址的 ARP 包以及用于域名解析的 DNS 包，wireshark 还捕获到若干 TCP 包。这些 TCP 包均为扫描机和靶机之间交换的，每次由扫描机向靶机各个端口发送 TCP 建立连接请求。

(1) 若靶机此端口未开放，则回复给扫描机[RST,ACK]连接失败。如下图所示，靶机 443 端口关闭。

4	0.031675653	192.168.204.134	192.168.204.135	TCP	74	36866 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=...
7	0.033396544	192.168.204.135	192.168.204.134	TCP	60	443 → 36866 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
10	0.056199338	192.168.204.134	192.168.204.135	TCP	74	58160 → 25 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=...
11	0.057578431	192.168.204.135	192.168.204.134	TCP	60	25 → 58160 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
12	0.057815131	192.168.204.134	192.168.204.135	TCP	74	35046 → 113 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=...
13	0.058548227	192.168.204.134	192.168.204.135	TCP	74	33136 → 135 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=...
14	0.059922121	192.168.204.135	192.168.204.134	TCP	60	113 → 35046 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
15	0.059959821	192.168.204.135	192.168.204.134	TCP	78	135 → 33136 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 WS=1 T...

(2) 若靶机此端口已开放，则回复给扫描机[SYN,ACK]，再由扫描机发送给靶机[ACK]，完成三次握手建立 TCP 连接。如下图所示，扫描机 33136 端口成功和靶机 135 端口建立 TCP 连接，说明靶机 135 端口已开放。

14	0.059922121	192.168.204.135	192.168.204.134	TCP	60	113 → 35046 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
15	0.059959821	192.168.204.135	192.168.204.134	TCP	78	135 → 33136 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 WS=1 T...
16	0.060192420	192.168.204.134	192.168.204.135	TCP	66	33136 → 135 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=141207668 TSec...
17	0.061391914	192.168.204.134	192.168.204.135	TCP	74	58146 → 3389 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSva...
18	0.062405109	192.168.204.135	192.168.204.134	TCP	60	3389 → 58146 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0

同理如下两图所示，可以看到扫描机成功和 139 端口、445 端口建立 TCP 连接。

68	0.107536299	192.168.204.134	192.168.204.135	TCP	74 36862 → 1862 [SYN, Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=141214824 TSecr=0 Win=0 Len=0
69	0.107918897	192.168.204.135	192.168.204.134	TCP	78 80 → 60576 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 WS=1 T...
70	0.107942497	192.168.204.135	192.168.204.134	TCP	78 139 → 36956 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 WS=1 T...
71	0.107965897	192.168.204.135	192.168.204.134	TCP	60 143 → 48546 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
72	0.108075196	192.168.204.134	192.168.204.135	TCP	66 60576 → 80 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=141207716 TSecr=...
1789	0.762015292	192.168.204.135	192.168.204.134	TCP	60 44501 → 59476 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2030	1.203328971	192.168.204.135	192.168.204.134	TCP	78 445 → 45060 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 WS=1 T...
2066	7.221614334	192.168.204.135	192.168.204.134	TCP	66 445 → 45060 [FIN, ACK] Seq=134 Ack=170 Win=64072 Len=0 TSval=18064...

(3) 扫描结束时，靶机的 135 端口、139 端口和 445 端口都向扫描机发送 [FIN,ACK]，扫描机回复 [ACK]，再由靶机发送 [FIN,ACK]，扫描机最终发送 [ACK]，完成四次挥手，TCP 连接终止。

time	source	destination	protocol	length	info
2054	7.215211560	192.168.204.135	192.168.204.134	TCP	66 135 → 33140 [FIN, ACK] Seq=1 Ack=33 Win=64208 Len=0 TSval=180647 T...
2055	7.216421655	192.168.204.135	192.168.204.134	SMB	199 Negotiate Protocol Response
2056	7.216548654	192.168.204.134	192.168.204.135	TCP	66 45060 → 445 [ACK] Seq=169 Ack=134 Win=64128 Len=0 TSval=141214824 T...
2057	7.217430351	192.168.204.134	192.168.204.135	TCP	66 33140 → 135 [ACK] Seq=33 Ack=2 Win=64256 Len=0 TSval=141214825 T...
2058	7.219394243	192.168.204.134	192.168.204.135	TCP	66 60586 → 80 [FIN, ACK] Seq=19 Ack=363 Win=64128 Len=0 TSval=1412148...
2059	7.219726642	192.168.204.134	192.168.204.135	TCP	66 33140 → 135 [FIN, ACK] Seq=33 Ack=2 Win=64256 Len=0 TSval=14121482...
2060	7.220122240	192.168.204.134	192.168.204.135	TCP	74 37654 → 135 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=...
2061	7.220564638	192.168.204.134	192.168.204.135	TCP	66 45060 → 445 [FIN, ACK] Seq=169 Ack=134 Win=64128 Len=0 TSval=14121...
2062	7.220568938	192.168.204.135	192.168.204.134	TCP	66 80 → 60586 [ACK] Seq=363 Ack=20 Win=64222 Len=0 TSval=180647 TSecr=...
2063	7.220607138	192.168.204.135	192.168.204.134	TCP	66 135 → 33140 [ACK] Seq=2 Ack=34 Win=64208 Len=0 TSval=180647 TSecr=...
2064	7.220868937	192.168.204.134	192.168.204.135	TCP	66 36970 → 139 [FIN, ACK] Seq=19 Ack=7 Win=64256 Len=0 TSval=14121482...
2065	7.221576434	192.168.204.135	192.168.204.134	TCP	78 135 → 37654 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 WS=1 T...
2066	7.221614334	192.168.204.135	192.168.204.134	TCP	66 445 → 45060 [FIN, ACK] Seq=134 Ack=170 Win=64072 Len=0 TSval=18064...
2067	7.221652634	192.168.204.135	192.168.204.134	TCP	66 139 → 36970 [ACK] Seq=7 Ack=20 Win=64222 Len=0 TSval=180647 TSecr=...
2068	7.221798433	192.168.204.134	192.168.204.135	TCP	74 45250 → 139 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=...
2069	7.222045932	192.168.204.134	192.168.204.135	TCP	66 37654 → 135 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=141214829 TSecr=...
2070	7.222092932	192.168.204.134	192.168.204.135	TCP	66 45060 → 445 [ACK] Seq=170 Ack=135 Win=64128 Len=0 TSval=141214830 T...
2071	7.222689930	192.168.204.135	192.168.204.134	TCP	78 139 → 45250 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 WS=1 T...
2072	7.222692629	192.168.204.134	192.168.204.135	TCP	234 37654 → 135 [PSH, ACK] Seq=1 Ack=1 Win=64256 Len=168 TSval=1412148...
2073	7.222846829	192.168.204.134	192.168.204.135	TCP	66 45250 → 139 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=141214830 TSecr=...
2074	7.223324327	192.168.204.134	192.168.204.135	SMB	234 Negotiate Protocol Request
2075	7.223735725	192.168.204.135	192.168.204.134	DCERPC	90 Bind nak: call id: 1073809408, Fragment: Single reason: Protocol v...
2076	7.224005024	192.168.204.134	192.168.204.135	TCP	66 37654 → 135 [ACK] Seq=169 Ack=25 Win=64256 Len=0 TSval=141214831 T...
2077	7.224510822	192.168.204.135	192.168.204.134	NBSS	71 Negative session response, Unspecified error
2078	7.224548621	192.168.204.135	192.168.204.134	TCP	66 135 → 37654 [FIN, ACK] Seq=25 Ack=169 Win=64072 Len=0 TSval=180647 T...
2079	7.224708821	192.168.204.134	192.168.204.135	TCP	66 37654 → 135 [FIN, ACK] Seq=169 Ack=25 Win=64256 Len=0 TSval=141214...
2080	7.225392218	192.168.204.134	192.168.204.135	TCP	66 37654 → 135 [ACK] Seq=170 Ack=26 Win=64256 Len=0 TSval=141214833 T...
2081	7.225823416	192.168.204.135	192.168.204.134	TCP	66 135 → 37654 [ACK] Seq=26 Ack=170 Win=64072 Len=0 TSval=180647 TSecr=...
2082	7.225897416	192.168.204.134	192.168.204.135	TCP	66 45250 → 139 [FIN, ACK] Seq=169 Ack=7 Win=64256 Len=0 TSval=1412148...
2083	7.227501200	192.168.204.135	192.168.204.134	TCP	66 139 → 45250 [ACK] Seq=7 Ack=170 Win=64072 Len=0 TSval=180647 TSecr=...

6.由以上分析可知，Nmap -sV 端口扫描的原理是由扫描机向靶机各端口发送 TCP 建立连接请求，若经历三次握手建立连接则说明此端口开启，最终通过四次挥手终止连接，否则说明此端口关闭。

3.4.2 -sS: TCP SYN 扫描，半开扫描

1.在扫描机 kali 终端输入 **nmap -sS 192.168.204.135** 命令对靶机端口进行 TCP SYN 扫描。终端输出的扫描结果如下图所示，可知靶机 135 端口、139 端口、445 端口开启，并列出了端口对应的版本信息。

```
(kali㉿kali)-[~]
└─$ sudo nmap -sS 192.168.204.135
[sudo] password for kali:
Starting Nmap 7.92 ( https://nmap.org ) at 2025-10-30 02:45 EDT
Nmap scan report for 192.168.204.135
Host is up (0.00094s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
MAC Address: 00:0C:29:5D:E8:C3 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.92 seconds
```

2.在上述 TCP SYN 端口扫描的同时，使用 wireshark 捕获网络流量，捕获结果如下图所示。

Time	Source	Destination	Protocol	Length	Info
1977 0.927405806	192.168.204.134	192.168.204.135	TCP	58	49199 → 10009 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1978 0.927613405	192.168.204.134	192.168.204.135	TCP	58	49199 → 9593 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1979 0.927751205	192.168.204.135	192.168.204.134	TCP	60	2910 → 49199 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1980 0.927843705	192.168.204.135	192.168.204.134	TCP	60	16113 → 49199 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1981 0.927966405	192.168.204.134	192.168.204.135	TCP	58	49199 → 631 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1982 0.928182505	192.168.204.134	192.168.204.135	TCP	58	49199 → 544 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1983 0.928249505	192.168.204.135	192.168.204.134	TCP	60	10009 → 49199 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1984 0.928286705	192.168.204.135	192.168.204.134	TCP	60	9593 → 49199 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1985 0.928389204	192.168.204.134	192.168.204.135	TCP	58	49199 → 2161 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1986 0.928595704	192.168.204.134	192.168.204.135	TCP	58	49199 → 16016 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1987 0.928679904	192.168.204.135	192.168.204.134	TCP	60	631 → 49199 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1988 0.928803004	192.168.204.134	192.168.204.135	TCP	58	49199 → 7007 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1989 0.929009704	192.168.204.134	192.168.204.135	TCP	58	49199 → 5414 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1990 0.929077504	192.168.204.135	192.168.204.134	TCP	60	544 → 49199 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1991 0.929115503	192.168.204.135	192.168.204.134	TCP	60	2161 → 49199 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1992 0.929153703	192.168.204.135	192.168.204.134	TCP	60	16016 → 49199 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1993 0.929292703	192.168.204.134	192.168.204.135	TCP	58	49199 → 705 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1994 0.929578003	192.168.204.134	192.168.204.135	TCP	58	49199 → 2020 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1995 0.929569703	192.168.204.135	192.168.204.134	TCP	60	7007 → 49199 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1996 0.929977403	192.168.204.135	192.168.204.134	TCP	60	5414 → 49199 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1997 0.930248102	192.168.204.134	192.168.204.135	TCP	58	49199 → 40193 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1998 0.930417202	192.168.204.135	192.168.204.134	TCP	60	705 → 49199 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1999 0.930540802	192.168.204.134	192.168.204.135	TCP	58	49199 → 1106 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
2000 0.930749901	192.168.204.134	192.168.204.135	TCP	58	49199 → 49161 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
2001 0.930881201	192.168.204.135	192.168.204.134	TCP	60	2020 → 49199 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2002 0.930919101	192.168.204.135	192.168.204.134	TCP	60	40193 → 49199 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2003 0.930959201	192.168.204.134	192.168.204.135	TCP	58	49199 → 43 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
2004 0.931164901	192.168.204.134	192.168.204.135	TCP	58	49199 → 9535 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
2005 0.931313301	192.168.204.135	192.168.204.134	TCP	60	1106 → 49199 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2006 0.931351000	192.168.204.135	192.168.204.134	TCP	60	49161 → 49199 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0

3.由上图可知，捕获到了大量扫描机和靶机间交换的 TCP 包。扫描机向靶机各个端口发送 TCP 建立连接请求[SYN]。

（1）若端口关闭，则回复给扫描机 TCP 报文[RST,ACK]，连接失败说明靶机该端口关闭。如下图的 3306 端口

1434 0.755471526	192.168.204.134	192.168.204.135	TCP	58	49199 → 3306 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
129 0.166489083	192.168.204.134	192.168.204.135	TCP	58	49199 → 3301 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
14 0.148277208	192.168.204.134	192.168.204.135	TCP	58	49199 → 3306 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
924 0.516924531	192.168.204.134	192.168.204.135	TCP	58	49199 → 3322 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1128 0.598980526	192.168.204.134	192.168.204.135	TCP	58	49199 → 3323 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
494 0.329394572	192.168.204.134	192.168.204.135	TCP	58	49199 → 3324 [SYN] Seq=0 Win=1024 Len=0 MSS=1460

（2）若端口开放，则靶机回复给扫描机[SYN,ACK]，稍后由主机回复[RST]终止建立连接。如下图所示靶机 139 端口为例。

43 0.154057100	192.168.204.134	192.168.204.135	TCP	58	49199 → 135 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
44 0.154120999	192.168.204.134	192.168.204.135	TCP	58	49199 → 23 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
45 0.154185799	192.168.204.134	192.168.204.135	TCP	58	49199 → 25 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
46 0.154349799	192.168.204.135	192.168.204.134	TCP	60	1720 → 49199 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
47 0.154363599	192.168.204.135	192.168.204.134	TCP	60	139 → 49199 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
48 0.154377199	192.168.204.135	192.168.204.134	TCP	60	143 → 49199 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
49 0.154390499	192.168.204.135	192.168.204.134	TCP	60	110 → 49199 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
50 0.154422799	192.168.204.134	192.168.204.135	TCP	54	49199 → 139 [RST] Seq=1 Win=0 Len=0
51 0.154605299	192.168.204.135	192.168.204.134	TCP	60	80 → 49199 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
52 0.154618699	192.168.204.135	192.168.204.134	TCP	60	135 → 49199 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
53 0.154632099	192.168.204.135	192.168.204.134	TCP	60	23 → 49199 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
54 0.154645499	192.168.204.135	192.168.204.134	TCP	60	25 → 49199 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
55 0.154668099	192.168.204.134	192.168.204.135	TCP	54	49199 → 80 [RST] Seq=1 Win=0 Len=0
56 0.154714099	192.168.204.134	192.168.204.135	TCP	54	49199 → 135 [RST] Seq=1 Win=0 Len=0
57 0.154890298	192.168.204.134	192.168.204.135	TCP	58	49199 → 995 [SYN] Seq=0 Win=1024 Len=0 MSS=1460

4.由以上分析可知，Nmap -sS 端口扫描（TCP SYN 扫描，半开扫描）由扫描机向靶机各个端口发送 TCP 建立连接请求，若靶机端口回复[SYN,ACK]则说明该端口开启，扫描机回复[RST]终止连接，并不完整地建立 TCP 连接。

5.将两种端口扫描进行比较，-sV 端口扫描耗时 7.74 秒，-sS 端口扫描耗时仅 1.42 秒，可知由于 TCP SYN 端口扫描未建立完整的 TCP 连接，扫描速度更快、更加省时高效；而-sV 模式由于建立了完整的 TCP 连接，相对更加完整可靠。

3.5 步骤五：操作系统扫描

1.在扫描机终端输入 **sudo su** 进入超级用户权限，在 **root** 权限下使用 **nmap -O 192.168.204.135** 命令启用操作系统检测，对靶机进行操作系统扫描。

```
(kali㉿kali)-[~]
└─$ sudo nmap -O 192.168.204.135
Starting Nmap 7.92 ( https://nmap.org ) at 2025-10-30 02:47 EDT
Nmap scan report for 192.168.204.135
Host is up (0.0014s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
MAC Address: 00:0C:29:5D:E8:C3 (VMware)
Device type: general purpose
Running: Microsoft Windows XP
OS CPE: cpe:/o:microsoft:windows_xp::sp2 cpe:/o:microsoft:windows_xp::sp3
OS details: Microsoft Windows XP SP2 or SP3
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 3.26 seconds
```

2.由上图终端输出的扫描结果可知，操作系统扫描得出靶机的开放端口分别是 135、139、445 端口，并输出其 Mac 地址、设置类型、操作系统类型及其 CPE 信息、操作系统细节，并说明靶机与扫描机的网络距离为 1 跳。

3.在进行上述操作系统扫描的同时，在扫描机中打开 **wireshark** 对网络流量进行捕获，捕获结果如下图所示。

No.	Time	Source	Destination	Protocol	Length	Info
1932	3.631813566	192.168.204.134	192.168.204.135	TCP	58	48982 → 9110 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1933	3.632046066	192.168.204.134	192.168.204.135	TCP	58	48982 → 10099 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1934	3.632264165	192.168.204.134	192.168.204.135	TCP	58	48982 → 3031 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1935	3.632199565	192.168.204.135	192.168.204.134	TCP	60	52673 → 48982 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1936	3.632237765	192.168.204.135	192.168.204.134	TCP	60	18101 → 48982 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1937	3.632275965	192.168.204.135	192.168.204.134	TCP	60	513 → 48982 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1938	3.63222765	192.168.204.135	192.168.204.134	TCP	60	8994 → 48982 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1939	3.63243085	192.168.204.134	192.168.204.135	TCP	58	48982 → 13450 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1940	3.632654964	192.168.204.134	192.168.204.135	TCP	58	48982 → 1031 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1941	3.632833964	192.168.204.134	192.168.204.135	TCP	58	48982 → 49158 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1942	3.633048064	192.168.204.134	192.168.204.135	TCP	58	48982 → 25735 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1943	3.633224663	192.168.204.134	192.168.204.135	TCP	58	48982 → 49176 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1944	3.633504763	192.168.204.134	192.168.204.135	TCP	58	48982 → 2065 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1945	3.633687063	192.168.204.134	192.168.204.135	TCP	58	48982 → 5988 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1946	3.633903862	192.168.204.134	192.168.204.135	TCP	58	48982 → 2135 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1947	3.633929262	192.168.204.135	192.168.204.134	TCP	60	9110 → 48982 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1948	3.633864662	192.168.204.135	192.168.204.134	TCP	60	10099 → 48982 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1949	3.633902162	192.168.204.135	192.168.204.134	TCP	60	3031 → 48982 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1950	3.633939462	192.168.204.135	192.168.204.134	TCP	60	13456 → 48982 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1951	3.633976662	192.168.204.135	192.168.204.134	TCP	60	1031 → 48982 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1952	3.634013962	192.168.204.135	192.168.204.134	TCP	60	49158 → 48982 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1953	3.634131862	192.168.204.134	192.168.204.135	TCP	58	48982 → 1087 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1954	3.634353461	192.168.204.134	192.168.204.135	TCP	58	48982 → 52848 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1955	3.634530661	192.168.204.134	192.168.204.135	TCP	58	48982 → 5950 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1956	3.634422861	192.168.204.135	192.168.204.134	TCP	60	25735 → 48982 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1957	3.634460761	192.168.204.135	192.168.204.134	TCP	60	49176 → 48982 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1958	3.634497661	192.168.204.135	192.168.204.134	TCP	60	2065 → 48982 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1959	3.634534661	192.168.204.135	192.168.204.134	TCP	60	5988 → 48982 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1960	3.634752660	192.168.204.134	192.168.204.135	TCP	58	48982 → 5907 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
2011	3.830620135	192.168.204.135	192.168.204.134	TCP	60	8073 → 48982 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2012	3.837982556	192.168.204.134	192.168.204.135	TCP	74	47263 → 80 [SYN] Seq=0 Win=1 Len=0 WS=1024 MSS=1460 TSval=42949672...
2013	3.839956452	192.168.204.135	192.168.204.134	TCP	78	80 → 47263 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 WS=1 TS...
2014	3.840197662	192.168.204.134	192.168.204.135	TCP	54	47263 → 80 [RST] Seq=1 Win=0 Len=0
2015	3.938700956	192.168.204.134	192.168.204.135	TCP	74	47264 → 80 [SYN] Seq=0 Win=63 Len=0 MSS=1460 WS=1 SACK_PERM=1 TSva...
2016	3.941205651	192.168.204.135	192.168.204.134	TCP	78	80 → 47264 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 WS=1 TS...
2017	3.941438151	192.168.204.134	192.168.204.135	TCP	54	47264 → 80 [RST] Seq=1 Win=0 Len=0
2018	4.039768754	192.168.204.134	192.168.204.135	TCP	74	47265 → 80 [SYN] Seq=0 Win=4 Len=0 TSval=4294967295 TSecr=0 WS=32 ...
2019	4.041235551	192.168.204.135	192.168.204.134	TCP	74	80 → 47265 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 WS=1 TS...
2020	4.041401951	192.168.204.134	192.168.204.135	TCP	54	47265 → 80 [RST] Seq=1 Win=0 Len=0
2021	4.140777353	192.168.204.134	192.168.204.135	TCP	70	47266 → 80 [SYN] Seq=0 Win=4 Len=0 SACK_PERM=1 TSval=4294967295 TS...
2022	4.142514850	192.168.204.135	192.168.204.134	TCP	78	80 → 47266 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 WS=1 TS...
2023	4.142715249	192.168.204.134	192.168.204.135	TCP	54	47266 → 80 [RST] Seq=1 Win=0 Len=0
2024	4.241198852	192.168.204.134	192.168.204.135	TCP	74	47267 → 80 [SYN] Seq=0 Win=10 Len=0 MSS=536 SACK_PERM=1 TSval=4294...
2025	4.242389050	192.168.204.135	192.168.204.134	TCP	78	80 → 47267 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 WS=1 TS...
2026	4.242598850	192.168.204.134	192.168.204.135	TCP	54	47267 → 80 [RST] Seq=1 Win=0 Len=0
2027	4.342965452	192.168.204.134	192.168.204.135	TCP	70	47268 → 80 [SYN] Seq=0 Win=512 Len=0 MSS=265 SACK_PERM=1 TSval=429...
2028	4.343502951	192.168.204.135	192.168.204.134	TCP	74	80 → 47268 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 TSval=0...
2029	4.344426450	192.168.204.134	192.168.204.135	TCP	54	47268 → 80 [RST] Seq=1 Win=0 Len=0
2030	4.368633802	192.168.204.134	192.168.204.135	ICMP	162	Echo (ping) request id=0x5c02, seq=295/9985, ttl=40 (no response ...
2031	4.370277749	192.168.204.135	192.168.204.134	ICMP	162	Echo (ping) reply id=0x5c02, seq=295/9985, ttl=128
2032	4.394307652	192.168.204.134	192.168.204.135	ICMP	192	Echo (ping) request id=0x5c03, seq=296/10241, ttl=54 (reply in 20...
2033	4.395390049	192.168.204.135	192.168.204.134	ICMP	192	Echo (ping) reply id=0x5c03, seq=296/10241, ttl=128 (request in...
2034	4.419933901	192.168.204.134	192.168.204.135	UDP	342	47335 → 32652 Len=300
2035	4.421767998	192.168.204.134	192.168.204.135	ICMP	190	Destination unreachable (Port unreachable)

4.由上图捕获结果可知，除了进行主机发现的 ARP 包和与网关进行域名解析的 DNS 包，wireshark 还捕获到若干 TCP 包、一个 UDP 包和几个 ICMP 包。

(1) TCP 包：在扫描机 192.168.204.134 和靶机 192.168.204.135 之间交换。

a) 建立 TCP 连接请求部分与 TCP SYN 半开端口扫描同理，由扫描机向靶机各

个端口发送[SYN]请求，若靶机该端口开放则回复[SYN,ACK]，再由扫描机回复[RST]拒绝 TCP 连接；若靶机该端口关闭则回复[RST,ACK]，建立 TCP 连接失败。

1743	3.578667472	192.168.204.135	192.168.204.134	TCP	60 9929 → 48982 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1744	3.578856672	192.168.204.134	192.168.204.135	TCP	58 48982 → 5357 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1745	3.579063171	192.168.204.134	192.168.204.135	TCP	58 48982 → 3995 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1746	3.579096471	192.168.204.135	192.168.204.134	TCP	60 6543 → 48982 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1747	3.579279671	192.168.204.134	192.168.204.135	TCP	58 48982 → 2009 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1748	3.579486371	192.168.204.134	192.168.204.135	TCP	58 48982 → 9081 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1749	3.579491771	192.168.204.135	192.168.204.134	TCP	60 3022 → 48982 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1750	3.579826070	192.168.204.135	192.168.204.134	TCP	60 5357 → 48982 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1751	3.579965270	192.168.204.134	192.168.204.135	TCP	58 48982 → 50000 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1752	3.580173369	192.168.204.134	192.168.204.135	TCP	58 48982 → 1040 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1753	3.580242169	192.168.204.135	192.168.204.134	TCP	60 3995 → 48982 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1754	3.580279169	192.168.204.135	192.168.204.134	TCP	60 2009 → 48982 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1755	3.580316169	192.168.204.135	192.168.204.134	TCP	60 9081 → 48982 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1756	3.580403569	192.168.204.134	192.168.204.135	TCP	58 48982 → 888 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1757	3.580606068	192.168.204.134	192.168.204.135	TCP	58 48982 → 2007 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1758	3.580743668	192.168.204.135	192.168.204.134	TCP	60 50000 → 48982 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1759	3.580809468	192.168.204.135	192.168.204.134	TCP	60 1040 → 48982 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1760	3.580820068	192.168.204.134	192.168.204.135	TCP	58 48982 → 65000 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1761	3.581022568	192.168.204.134	192.168.204.135	TCP	58 48982 → 9917 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1762	3.581226567	192.168.204.134	192.168.204.135	TCP	58 48982 → 10025 [SYN] Seq=0 Win=1024 Len=0 MSS=1460

b) 在对靶机端口全部扫描完毕后，扫描机向已知开放的靶机 139 端口再次发送 [SYN]，收到[SYN,ACK]回复后向其回复[SYN]经历三次握手成功建立 TCP 连接。随后扫描机又向靶机 139 端口发送了 [SYN,ECN,CWR,Reserved]，表示拥塞通知，告诉靶机 139 端口减小拥塞窗口，在靶机回复[SYN,ACK]后扫描机仍回复[RST]拒绝连接。

28	3.590730050	192.168.204.135	192.168.204.134	TCP	60 22 → 48982 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
29	3.590835949	192.168.204.134	192.168.204.135	TCP	54 48982 → 135 [RST] Seq=1 Win=0 Len=0
30	3.591012749	192.168.204.134	192.168.204.135	TCP	54 48982 → 80 [RST] Seq=1 Win=0 Len=0
31	3.591218248	192.168.204.135	192.168.204.134	TCP	60 1720 → 48982 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
32	3.592455546	192.168.204.135	192.168.204.134	TCP	60 441 → 48982 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
33	3.592493946	192.168.204.135	192.168.204.134	TCP	60 139 → 48982 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
34	3.592576246	192.168.204.134	192.168.204.135	TCP	54 48982 → 139 [RST] Seq=1 Win=0 Len=0
35	3.595070941	192.168.204.134	192.168.204.135	TCP	58 48982 → 21 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
36	3.595365140	192.168.204.134	192.168.204.135	TCP	58 48982 → 256 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
37	3.595844939	192.168.204.134	192.168.204.135	TCP	58 48982 → 445 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
38	3.596071739	192.168.204.134	192.168.204.135	TCP	58 48982 → 443 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
39	3.596279739	192.168.204.134	192.168.204.135	TCP	58 48982 → 8080 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
40	3.596406738	192.168.204.134	192.168.204.135	TCP	58 48982 → 143 [SYN] Seq=0 Win=1024 Len=0 MSS=1460

(2) UDP 包：在对靶机端口半开扫描结束后，扫描机发送 UDP 报文到靶机主机 32652 端口，靶机随后回复 ICMP 包显示端口不可达。

2033	4.395390049	192.168.204.135	192.168.204.134	ICMP	192 Echo (ping) reply id=0x5c03, seq=296/10241, ttl=128 (request in...
2034	4.419933901	192.168.204.134	192.168.204.135	UDP	342 47335 → 32652 Len=300
2035	4.421707098	192.168.204.135	192.168.204.134	ICMP	190 Destination unreachable (Port unreachable)
2036	4.445842350	192.168.204.134	192.168.204.135	TCP	66 47275 → 80 [SYN, ECN, CWR, Reserved] Seq=0 Win=3 Len=0 WS=1024 MSS...
2037	4.448463645	192.168.204.135	192.168.204.134	TCP	66 80 → 47275 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 WS=1 SA...
2038	4.448717144	192.168.204.134	192.168.204.135	TCP	54 47275 → 80 [RST] Seq=1 Win=0 Len=0

(3) ICMP 包：利用 ping 命令验证网络的连通性并统计响应时间和 TTL (IP 包生存周期)。

2033	4.395390049	192.168.204.135	192.168.204.134	ICMP	192 Echo (ping) reply id=0x5c03, seq=296/10241, ttl=128 (request in...
2034	4.419933901	192.168.204.134	192.168.204.135	UDP	342 47335 → 32652 Len=300
2035	4.421707098	192.168.204.135	192.168.204.134	ICMP	190 Destination unreachable (Port unreachable)
2036	4.445842350	192.168.204.134	192.168.204.135	TCP	66 47275 → 80 [SYN, ECN, CWR, Reserved] Seq=0 Win=3 Len=0 WS=1024 MSS...
2037	4.448463645	192.168.204.135	192.168.204.134	TCP	66 80 → 47275 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 WS=1 SA...
2038	4.448717144	192.168.204.134	192.168.204.135	TCP	54 47275 → 80 [RST] Seq=1 Win=0 Len=0

5.由以上分析可知，Nmap 操作系统扫描通过发送一系列 TCP 和 UDP 报文到远程主机并检查响应中的每一个比特，将其与 TCP/IP 协议栈 fingerprinting 数据库中的已知操作系统进行比较，打印匹配出的操作系统详细信息即可告知用户靶机操作系统的详细信息。

3.6 步骤六：漏洞扫描

1.在扫描机 kali 终端使用 **nmap 192.168.204.135 -script=vuln** 命令进行靶机常见漏洞扫描，终端输出的扫描结果如下三图所示。

```
(kali㉿kali)-[~]
└─$ nmap 192.168.204.135 -script=vuln
Starting Nmap 7.92 ( https://nmap.org ) at 2025-10-30 02:47 EDT
Nmap scan report for 192.168.204.135
Host is up (0.0020s latency).
Not shown: 997 closed tcp ports (conn-refused)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds

Host script results:
|_ smb-vuln-ms08-067:
|   VULNERABLE:
|     Microsoft Windows system vulnerable to remote code execution (MS08-067)
|     State: VULNERABLE
|     IDs: CVE:CVE-2008-4250
|     The Server service in Microsoft Windows 2000 SP4, XP SP2 and SP3, Server 2003 SP1 and SP2, Vista Gold and SP1, Server 2008, and 7 Pre-Beta allows remote attackers to execute a arbitrary code via a crafted RPC request that triggers the overflow during path canonicalization.
|     Disclosure date: 2008-10-23
|     References:
|       https://technet.microsoft.com/en-us/library/security/ms08-067.aspx
|       https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2008-4250
|_ smb-vuln-cve-2012-1182: NT_STATUS_ACCESS_DENIED
|_ smb-vuln-ms10-054: false
|_ smb-vuln-ms10-061: ERROR: Script execution failed (use -d to debug)
|_ smb-vuln-ms17-010:
|   VULNERABLE:
|     Remote Code Execution vulnerability in Microsoft SMBv1 servers (ms17-010)
|     State: VULNERABLE
|     IDs: CVE:CVE-2017-0143
|     Risk factor: HIGH
|     A critical remote code execution vulnerability exists in Microsoft SMBv1 servers (ms17-010).
|     Disclosure date: 2017-03-14
|     References:
|       https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2017-0143
|       https://technet.microsoft.com/en-us/library/security/ms17-010.aspx
|       https://blogs.technet.microsoft.com/msrc/2017/05/12/customer-guidance-for-wannacrypt-attacks/

Nmap done: 1 IP address (1 host up) scanned in 39.85 seconds
```

2.在以上扫描过程中，打开 wireshark 对网络流量进行捕获，捕获结果如下两图所示。

o.	Time	Source	Destination	Protocol	Length	Info
2120	11.894175687	192.168.204.134	192.168.204.135	TCP	66	55894 → 80 [ACK] Seq=19 Ack=363 Win=64128 Len=0 TSval=144459705 TS...
2121	11.894473287	192.168.204.134	192.168.204.135	TCP	66	55884 → 80 [ACK] Seq=19 Ack=363 Win=64128 Len=0 TSval=144459705 TS...
2122	11.899314688	192.168.204.134	192.168.204.135	TCP	74	45044 → 445 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=...
2123	11.899659788	192.168.204.134	192.168.204.135	TCP	66	55884 → 80 [FIN, ACK] Seq=19 Ack=363 Win=64128 Len=0 TSval=1444597...
2124	11.901118689	192.168.204.135	192.168.204.134	TCP	78	445 → 45044 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 WS=1 T...
2125	11.901362289	192.168.204.134	192.168.204.135	TCP	66	45044 → 445 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=144459712 TSecr...
2126	11.901139089	192.168.204.135	192.168.204.134	TCP	66	80 → 55884 [ACK] Seq=363 Ack=20 Win=64222 Len=0 TSval=213424 TSecr...
2127	11.951384904	192.168.204.134	192.168.204.135	TCP	74	45054 → 445 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=...
2128	11.951843404	192.168.204.134	192.168.204.135	TCP	66	55894 → 80 [FIN, ACK] Seq=19 Ack=363 Win=64128 Len=0 TSval=1444597...
2129	11.952359504	192.168.204.135	192.168.204.134	TCP	78	445 → 45054 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 WS=1 T...
2130	11.952464404	192.168.204.134	192.168.204.135	TCP	66	45054 → 445 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=144459763 TSecr...
2131	11.952381104	192.168.204.135	192.168.204.134	TCP	66	80 → 55894 [ACK] Seq=363 Ack=20 Win=64222 Len=0 TSval=213424 TSecr...
2132	11.952890104	192.168.204.134	192.168.204.135	TCP	74	55900 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=...
2133	11.953422604	192.168.204.135	192.168.204.134	TCP	78	80 → 55900 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 WS=1 TS...
2134	11.953670104	192.168.204.134	192.168.204.135	TCP	66	55900 → 80 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=144459764 TSecr...
2135	11.954598905	192.168.204.134	192.168.204.135	TCP	74	55910 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=...
2136	11.955495505	192.168.204.135	192.168.204.134	TCP	78	80 → 55910 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 WS=1 TS...
2137	11.955767005	192.168.204.134	192.168.204.135	TCP	66	55910 → 80 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=144459766 TSecr...
2138	11.956683205	192.168.204.134	192.168.204.135	TCP	74	55912 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=...
2139	11.957020905	192.168.204.134	192.168.204.135	SMB	119	Negotiate Protocol Request
2140	11.957445606	192.168.204.135	192.168.204.134	TCP	78	80 → 55912 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 WS=1 TS...
2141	11.957524005	192.168.204.134	192.168.204.135	TCP	66	55912 → 80 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=144459768 TSecr...
2142	11.957974906	192.168.204.135	192.168.204.134	SMB	199	Negotiate Protocol Response
2143	11.958020206	192.168.204.134	192.168.204.135	TCP	66	45044 → 445 [ACK] Seq=54 Ack=134 Win=64128 Len=0 TSval=144459769 T...
2144	12.007583020	192.168.204.135	192.168.204.134	TCP	66	135 → 51008 [ACK] Seq=1 Ack=111 Win=64130 Len=0 TSval=213426 TSecr...
2145	12.007738920	192.168.204.134	192.168.204.135	TCP	74	45056 → 445 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=...
2146	12.008486320	192.168.204.134	192.168.204.135	SMB	119	Negotiate Protocol Request
2147	12.008780521	192.168.204.135	192.168.204.134	TCP	78	445 → 45056 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 WS=1 T...
2148	12.008870720	192.168.204.134	192.168.204.135	TCP	66	45056 → 445 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=144459820 TSecr...

3.由图中捕获结果可知，除了进行主机发现的 ARP 包和与网关间进行域名解析的 DNS 包，wireshark 还捕获到大量 TCP 包、SSLv3 包、SMB 包和 NBNS 包。

(1) TCP 包：在扫描机 192.168.204.134 和靶机 192.168.204.135 之间交换数据，扫描机向靶机各端口发送 TCP 连接建立请求[SYN]，若靶机该端口开放则回复[SYN,ACK]，再由扫描机回复[ACK]完成三次握手，成功与此端口建立 TCP 连接；若该端口未开放则回复[RST,ACK]建立 TCP 连接失败。

1954	11.680236224	192.168.204.135	192.168.204.134	TCP	60 27352 → 36909 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1955	11.680255124	192.168.204.135	192.168.204.134	TCP	60 1687 → 36909 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1956	11.680275024	192.168.204.135	192.168.204.134	TCP	60 1666 → 36909 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1957	11.680548424	192.168.204.134	192.168.204.135	TCP	58 36909 → 8099 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1958	11.680829624	192.168.204.134	192.168.204.135	TCP	58 36909 → 6969 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1959	11.680923824	192.168.204.134	192.168.204.135	TCP	58 36909 → 16001 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1960	11.681049624	192.168.204.134	192.168.204.135	TCP	58 36909 → 16012 [SYN] Seq=0 Win=1024 Len=0 MSS=1460

(2) SSLv3 包、SMB 包和 NBNS 包：首先由扫描机向靶机发送内容未 Client Hello 的 SSLv3 包建立连接；随后由扫描机向靶机发送 NBNS 包请求 NBSTAT 端口查询服务，靶机收到后发送 NBNS 包回复 NBSTAT 服务开启；之后由扫描机向靶机发送 SMB negotiate protocol request 请求数据报并列出它所支持的所有 SMB 协议版本，靶机收到请求信息后响应请求并列出希望使用的协议版本。

2195	12.271810998	192.168.204.134	192.168.204.135	SMB	142 Tree Connect AndX Request, Path: \\192.168.204.135\IPC\$
2192	12.222226583	192.168.204.135	192.168.204.134	SMB	199 Negotiate Protocol Response
2191	12.221085282	192.168.204.134	192.168.204.135	SMB	119 Negotiate Protocol Request
2182	12.165337066	192.168.204.135	192.168.204.134	SMB	155 Negotiate Protocol Response
2180	12.164697666	192.168.204.135	192.168.204.134	SMB	158 Session Setup AndX Response
2179	12.164496066	192.168.204.134	192.168.204.135	SMB	119 Negotiate Protocol Request
2178	12.164092966	192.168.204.134	192.168.204.135	SMB	153 Session Setup AndX Response, Error: STATUS_LOGON_FAILURE
2171	12.112718351	192.168.204.135	192.168.204.134	SMB	155 Negotiate Protocol Response
2168	12.111266751	192.168.204.134	192.168.204.135	SMB	119 Negotiate Protocol Request
2165	12.062656736	192.168.204.135	192.168.204.134	SMB	105 Session Setup AndX Response, Error: STATUS_LOGON_FAILURE
2163	12.062299236	192.168.204.135	192.168.204.134	SMB	155 Negotiate Protocol Response
2162	12.060363836	192.168.204.134	192.168.204.135	SMB	119 Negotiate Protocol Request
2159	12.059824435	192.168.204.134	192.168.204.135	SMB	205 Session Setup AndX Request, User: \guest
2150	12.009346721	192.168.204.135	192.168.204.134	SMB	155 Negotiate Protocol Response
2146	12.008486320	192.168.204.134	192.168.204.135	SMB	119 Negotiate Protocol Request
2142	11.957974906	192.168.204.135	192.168.204.134	SMB	199 Negotiate Protocol Response
2139	11.957020905	192.168.204.134	192.168.204.135	SMB	119 Negotiate Protocol Request
2107	11.947550372	192.168.204.134	192.168.204.135	SMB	144 Negotiate Protocol Request

(3) 在收到靶机回复前，扫描机向靶机 445 端口发送 TCP 连接建立请求 [SYN]，收到 445 端口回复的[SYN.ACK]后向靶机回复[ACK]成功建立 TCP 连接进行通信。

4.由以上分析可知，Nmap 漏洞扫描是扫描机首先通过向各端口发送 TCP 连接建立请求扫描靶机开放的端口，随后每次向靶机发送 SMB 报文并与开放的 445 端口建立 TCP 连接进行通信后，靶机回复 SMB 报文即为对相应漏洞的扫描结果。

3.7 步骤七：主机全面扫描

1.在扫描机 kali 终端输入 **nmap -T4 -A 192.168.204.135** 对靶机进行完整全面的扫描。其中-A 选项用于使用进攻性方式扫描；-T4 指定扫描过程使用的时序，总共 6 个级别（0-5），级别越高扫描速度越快，但也越易被防火墙/IDS 检测并屏蔽掉，在网络状况良好的情况推荐使用 T4。终端输出的扫描结果如下两图所示。

```
(kali㉿kali)-[~]
$ sudo nmap -T4 -A 192.168.204.135
Starting Nmap 7.92 ( https://nmap.org ) at 2025-10-30 02:49 EDT
Nmap scan report for 192.168.204.135
Host is up (0.0015s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
135/tcp    open  msrpc          Microsoft Windows RPC
139/tcp    open  netbios-ssn    Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds   Windows XP microsoft-ds
MAC Address: 00:0C:29:5D:E8:C3 (VMware)
Device type: general purpose
Running: Microsoft Windows XP
OS CPE: cpe:/o:microsoft:windows_xp::sp2 cpe:/o:microsoft:windows_xp::sp3
OS details: Microsoft Windows XP SP2 or SP3
Network Distance: 1 hop
Service Info: OSs: Windows, Windows XP; CPE: cpe:/o:microsoft:windows, cpe:/o:microsoft:windows_xp

Host script results:
|_ clock-skew: mean: -4h00m00s, deviation: 5h39m23s, median: -8h00m00s
|_ smb2-time: Protocol negotiation failed (SMB2)
|_ nbstat: NetBIOS name: WINXP-3D8CC5A43, NetBIOS user: <unknown>, NetBIOS MAC: 00:0c:29:5d:e8:c3 (VMware)
|_ smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_ message_signing: disabled (dangerous, but default)
|_ smb-os-discovery:
|   OS: Windows XP (Windows 2000 LAN Manager)
|   OS CPE: cpe:/o:microsoft:windows_xp::-
|   Computer name: winxp-3d8cc5a43
|   NetBIOS computer name: WINXP-3D8CC5A43\x00
|   Workgroup: WORKGROUP\x00
|_ System time: 2025-10-30T14:49:46+08:00

TRACEROUTE
HOP RTT      ADDRESS
1   1.55 ms  192.168.204.135

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 21.16 seconds
```

2.由扫描结果可知，全面扫描靶机得到主机发现的结果为“Host is up”；端口扫描的结果为共 997 个关闭端口、3 个开放端口分别为 135、139、445 端口，并列 出 开 放 端 口 的 对 应 服 务 和 版 本 信 息 ； 操 作 系 统 扫 描 的 结 果 为 靶 机 OS 为 windows XP 并输出其 CPE 信息等；版本侦测扫描的结果为对扫描到的开放状况进一步探测端口上运行的具体的应用程序和版本信息；并调用 NSE 脚本进行进一步的信息挖掘的显示结果。

3.在进行上述扫描的同时，打开 wireshark 对网络流量进行捕获，捕获结果如下图所示。

No.	Time	Source	Destination	Protocol	Length	Info
2366	11.324585182	192.168.204.135	192.168.204.134	TCP	66	80 → 47890 [ACK] Seq=319 Ack=160 Win=64082 Len=0 TSval=220053 TSecr=145...
2367	11.324649882	192.168.204.134	192.168.204.135	TCP	74	53674 → 445 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=1451...
2368	11.325572180	192.168.204.134	192.168.204.135	HTTP	219	GET / HTTP/1.1
2369	11.325894380	192.168.204.135	192.168.204.134	TCP	78	445 → 53674 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 WS=1 TSval=...
2370	11.326914860	192.168.204.134	192.168.204.135	SMB2	250	Negotiate Protocol Request
2371	11.326271979	192.168.204.134	192.168.204.135	TCP	66	53674 → 445 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=145122591 TSecr=0
2372	11.327858376	192.168.204.135	192.168.204.134	HTTP	427	HTTP/1.1 200 OK (text/html)
2373	11.327897476	192.168.204.135	192.168.204.134	TCP	66	80 → 47914 [FIN, ACK] Seq=362 Ack=154 Win=64087 Len=0 TSval=220053 TSecr=...
2374	11.328026276	192.168.204.134	192.168.204.135	TCP	66	47914 → 80 [ACK] Seq=154 Ack=362 Win=64128 Len=0 TSval=145122593 TSecr=...
2375	11.328448975	192.168.204.134	192.168.204.135	TCP	74	53680 → 445 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=1451...
2376	11.329335773	192.168.204.135	192.168.204.134	TCP	78	445 → 53680 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 WS=1 TSval=...
2377	11.329474073	192.168.204.134	192.168.204.135	TCP	66	47902 → 80 [FIN, ACK] Seq=164 Ack=319 Win=64128 Len=0 TSval=145122595 T...
2378	11.329704173	192.168.204.134	192.168.204.135	TCP	66	53680 → 445 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=145122595 TSecr=0
2379	11.330277771	192.168.204.134	192.168.204.135	HTTP	219	GET / HTTP/1.1
2380	11.330540671	192.168.204.134	192.168.204.135	HTTP	281	OPTIONS / HTTP/1.1
2381	11.330558171	192.168.204.135	192.168.204.134	TCP	66	80 → 47902 [ACK] Seq=319 Ack=165 Win=64077 Len=0 TSval=220053 TSecr=145...
2382	11.331099670	192.168.204.134	192.168.204.135	SMB	205	Session Setup AndX Request, User: \guest
2383	11.331218770	192.168.204.134	192.168.204.135	UDP	56	58753 → 21226 Len=14
2384	11.331739769	192.168.204.134	192.168.204.135	SMB2	250	Negotiate Protocol Request
2385	11.332060868	192.168.204.135	192.168.204.134	HTTP	427	HTTP/1.1 200 OK (text/html)
2386	11.332145368	192.168.204.135	192.168.204.134	ICMP	84	Destination unreachable (Port unreachable)
2387	11.332292968	192.168.204.134	192.168.204.135	TCP	66	47930 → 80 [ACK] Seq=154 Ack=362 Win=64128 Len=0 TSval=145122597 TSecr=...
2388	11.332946766	192.168.204.135	192.168.204.134	TCP	66	80 → 47930 [FIN, ACK] Seq=362 Ack=154 Win=64087 Len=0 TSval=220053 TSecr=...
2389	11.332984666	192.168.204.135	192.168.204.134	HTTP	389	HTTP/1.1 405 Not Allowed (text/html)
2390	11.333229866	192.168.204.134	192.168.204.135	TCP	66	47940 → 80 [ACK] Seq=216 Ack=324 Win=64128 Len=0 TSval=145122598 TSecr=...
2391	11.333770665	192.168.204.135	192.168.204.134	TCP	66	80 → 47940 [FIN, ACK] Seq=324 Ack=216 Win=64025 Len=0 TSval=220053 TSecr=...
2392	11.333810465	192.168.204.135	192.168.204.134	SMB	105	Session Setup AndX Response, Error: STATUS_LOGON_FAILURE
2393	11.334124264	192.168.204.134	192.168.204.135	TCP	66	53644 → 445 [ACK] Seq=193 Ack=173 Win=64128 Len=0 TSval=145122599 TSecr=...
2394	11.371957596	192.168.204.134	192.168.204.135	TCP	66	47914 → 80 [ACK] Seq=154 Ack=363 Win=64128 Len=0 TSval=145122637 TSecr=...
2395	11.376277887	192.168.204.134	192.168.204.135	TCP	66	47940 → 80 [ACK] Seq=216 Ack=325 Win=64128 Len=0 TSval=145122641 TSecr=...

▶ Frame 1: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface eth0, id 0
 ▶ Ethernet II, Src: VMware_e8:44:a5 (00:50:56:e8:44:a5), Dst: VMware_a5:82:6c (00:0c:29:a5:82:6c)

4.由上图捕获结果可知，wireshark 捕获到 ARP 包及 DNS 包还有大量 TCP 和 UDP 包、ICMP 包、SMB 包、SMB2 包及 NBSS 包。

- (1) ARP 包：主机发现。
- (2) TCP 包：端口扫描。
- (3) TCP 和 UDP 包、ICMP 包：操作系统扫描及版本侦测。
- (4) SMB 包、SMB2 包、NBSS 包：调用 NSE 脚本进行进一步的信息挖掘。

四、实验总结

本次实验使我得以将网络扫描技术从理论层面延伸至具体的动手实践，完成了一次对目标靶机的系统性探测与分析。我以 Nmap 为核心工具，展开了一条从宏观发现到深度探测的完整路径。实验始于基础的主机发现，通过 Ping 扫描快速定位了网络中的活跃设备；继而深入到端口扫描，对比体验了 TCP SYN 半开扫描的高效与 TCP Connect 全连接扫描的可靠性，并尝试了探测 UDP 服务这一更具挑战性的任务；随后，通过服务版本侦测和操作系统识别，将单纯的端口号转化为更具价值的情报信息；最终，借助 Nmap 强大的脚本引擎，我对靶机进行了初步的漏洞扫描，将之前获取的信息与已知的安全风险关联起来。

整个过程中，我同步使用 Wireshark 进行网络流量抓取与分析，这项工作本次实验的精华所在。当在终端中执行一条条 Nmap 命令时，Wireshark 如同一个高精度的“显微镜”，让我得以窥见命令背后真实的网络对话。例如，我清晰地观察到 TCP SYN 扫描如何通过发送 SYN 包、接收 SYN/ACK 包后即刻发送 RST 包来中断连接，从而验证了其“半开扫描”的高效原理。与之形成鲜明对比的是，TCP Connect 扫描则展示了完整的三次握手过程，这使我深刻理解了两种扫描方式在系统调用层级和隐蔽性上的根本差异。而对操作系统识别过程的抓包分析则更为有趣，我看到 Nmap 并非进行单一请求，而是发送一系列精心构造的、带有特殊 TCP 参数的探测包，通过分析目标系统独特的响应“指纹”来推断其操作系统。这种将扫描命令与底层网络流量相互印证的过程，极大地深化了我对

网络协议和扫描技术内在机制的理解。

经过这次亲自动手的实验，我所获得的远不止是几个命令的使用技巧。它让我真切地体会到，一次看似简单的扫描背后，是大量网络数据包的交互与博弈，信息安全的攻防在最底层便已悄然展开。这极大地提升了我对网络安全隐患的警觉性。我认识到，网络中开放的端口、未更新补丁的服务、甚至是系统本身透露的指纹信息，都可能成为攻击者利用的突破口。因此，作为一名安全实践者，必须时刻保持主动防御的意识，不仅要熟练掌握漏洞扫描这一关键技能，更要理解其原理以应对可能的规避手段，同时还要坚持定期更新漏洞库，不断提升扫描技术的深度与广度。唯有如此，才能在这场动态的攻防较量中，更有效地发现隐患、评估风险，从而切实提升网络的整体安全水位。